

CS201 Midterm 2

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures.

I. QUESTION (BINARY SEARCH TREE)

T1 and T2 are two binary trees. Write the recursive method

```
boolean isIdentical(TreeNode T1,
                    TreeNode T2)
```

in Tree or TreeNode class to determine if T1 is identical to T2.

II. QUESTION (BINARY SEARCH TREE)

Write the non-recursive method

```
int product()
```

in Tree class that computes the products of all keys in a binary search tree by using stack.

III. QUESTION (QUEUE)

Write another constructor method

```
void Queue(Queue[] list)
```

which constructs a new array based queue by adding the elements in the *list* of queues one by one. So, the first k elements of the original queue will be constructed with the first elements of the k queues in the list; the second k elements of the original queue will be constructed with the second elements of the k queues in the list etc. The elements from queues should be recreated (not copied from the queues). You are not allowed to use enqueue, dequeue, isEmpty functions. You should solve the question for array implementation.

IV. QUESTION (QUEUE)

Write the method

```
void removeAll(Queue[] list)
```

which removes all elements in the queues in the *list* from the original queue. You are not allowed to use enqueue, dequeue, isEmpty functions. You should solve the question for list implementation.

V. QUESTION (HASHING)

Write the static method in Hash class

```
boolean sumOfFourK(int[] array, int K)
```

that takes an array of integers as a parameter and returns true if the sum of four elements in the array is K. Your method should run in $O(N^2)$ time. You are only allowed to use an external array and an external hash table.

VI. QUESTION (HASHING)

Write a static method

```
int [] union(int[] list1, int[] list2)
```

to find the union of the elements in two arrays and return a new array. The union array should contain only that many items not more not less. Your method should run in $O(N)$ time, where N is the number of elements in the arrays. Do not use any external data structures or arrays except the resulting array and an external hash table.